

IVAN SESLIJA

SOFTWARE ENGINEER Python • JavaScript • React • AI/ML • Automation

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Portfolio • GitHub • LinkedIn

Computer Science graduate and software developer focused on practical engineering: web applications, automation, algorithmic systems, and machine learning. Built projects across React and REST APIs, PyTorch reinforcement learning, A* pathfinding, computer vision, and production-facing VBA automation.

LANGUAGES Python, JavaScript, Java, C, C++, Visual Basic/VBA	WEB & SOFTWARE React, Vite, REST APIs, YouTube Data API, Git/GitHub	AI / ML & TOOLS PyTorch, TensorFlow, scikit-learn, OpenCV, Gymnasium, NumPy, TensorBoard
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SELECTED PROJECTS

YouTube Random Finder <i>React • Vite • JavaScript • YouTube Data API v3</i> - Built a video-discovery app with debounced channel search, region/category filtering, configurable archive depth, and session duplicate prevention. - Structured the UI with reusable React components and custom hooks; persisted user preferences with localStorage.	Breakout DDQN <i>Python • PyTorch • Gymnasium • TensorBoard</i> - Implemented a Double Dueling DQN agent for Atari Breakout using CNNs, experience replay, target-network updates, and action-value decomposition. - Added checkpointing, training metrics, TensorBoard logging, and tools to watch/evaluate trained-agent gameplay.
PacMan AI <i>JavaScript • p5.js • A* • Reinforcement Learning</i> - Developed a grid-based Pac-Man environment with real-time rendering and A* pathfinding for dynamic ghost navigation. - Supported manual and AI controls and experimented with reward-driven reinforcement learning for Pac-Man behavior.	Data Mining & ML <i>Python • TensorFlow • scikit-learn • NumPy</i> - Built Fashion-MNIST neural-network classifiers with normalization, label-noise experiments, and activation-function comparisons. - Performed hyperparameter tuning and k-fold cross-validation; generated error plots to compare model configurations.
Peggle Master <i>Python • OpenCV • PyAutoGUI • PyGetWindow</i> - Built a computer-vision automation tool that detects game elements and programmatically controls mouse input to automate Peggle Deluxe gameplay. - Combined screen analysis with window/input automation to interact with a running desktop application.	Maze Generators <i>Python • NumPy • OpenCV • Matplotlib</i> - Implemented randomized depth-first-search maze generation using an explicit stack and visited-cell tracking. - Rendered generated mazes to images with OpenCV/Matplotlib and marked start/end locations for visualization.

EXPERIENCE

- Mercana Furniture and Decor | Customer Operations Coordinator** *Surrey, BC | Jan 2026 - Present*
- Develop Visual Basic/VBA automations to eliminate repetitive operational work; built an automated weekly art-report workflow that compiles and distributes information to sales representatives.
 - Provide hands-on IT support for computers, printers, software, hardware, and networking equipment while identifying workflow bottlenecks suited to technical solutions.
 - Prepare, configure, and deploy technology for employee setups and off-site company events, including computers, printers, and networking equipment.
- MXS ON SITE IT SERVICES INC | IT Field Worker** *Vancouver, BC | Aug 2018 - Aug 2022*
- Installed, configured, and troubleshoot computers, networking equipment, and electronic hardware; diagnosed connectivity issues and coordinated with remote technical teams on repairs, upgrades, and replacements.
 - Served as the on-site technical contact for service calls, translating issues between clients and remote teams while adapting to unexpected field problems.
- ADDITIONAL EXPERIENCE** Uber — Driver (Vancouver, 2024-2025) • Vitran Express — Warehouse Worker (Surrey, 2022) • Speedy Bob Logistics — Assistant Safety Manager (Langley, 2015-2016)

EDUCATION University of Victoria — B.E., Computer Science
2016 - 2023

LANGUAGES
English • Serbian • Macedonian • French • Spanish